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12.
13. # Every possible symbol that can be encrypted/decrypted:
14. # (!) You can add numbers and punctuation marks to encrypt those
15. # symbols as well.
16. SYMBOLS = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'
17.
18. print('Caesar Cipher, by Al Sweigart al@inventwithpython.com')
19. print('The Caesar cipher encrypts letters by shifting them over by a')
20. print('key number. For example, a key of 2 means the letter A is')
21. print('encrypted into C, the letter B encrypted into D, and so on.')
22. print()
23.
24. # Let the user enter if they are encrypting or decrypting:
25. while True: # Keep asking until the user enters e or d.
26.     print('Do you want to (e)ncrypt or (d)ecrypt?')
27.     response = input('> ').lower()
28.     if response.startswith('e'):
29.         mode = 'encrypt'
30.         break
31.     elif response.startswith('d'):
32.         mode = 'decrypt'
33.         break
34.     print('Please enter the letter e or d.')
35.
36. # Let the user enter the key to use:
37. while True: # Keep asking until the user enters a valid key.
38.     maxKey = len(SYMBOLS) - 1
39.     print('Please enter the key (0 to {}) to use.'.format(maxKey))
40.     response = input('> ').upper()
41.     if not response.isdecimal():
42.         continue
43.
44.     if 0 <= int(response) < len(SYMBOLS):
45.         key = int(response)
46.         break
47.
48. # Let the user enter the message to encrypt/decrypt:
49. print('Enter the message to {}'.format(mode))
50. message = input('> ')
51.
52. # Caesar cipher only works on uppercase letters:
53. message = message.upper()
54.
55. # Stores the encrypted/decrypted form of the message:
56. translated = ''
57.
58. # Encrypt/decrypt each symbol in the message:
59. for symbol in message:
60.     if symbol in SYMBOLS:
61.         # Get the encrypted (or decrypted) number for this symbol.
62.         num = SYMBOLS.find(symbol) # Get the number of the symbol.
63.         if mode == 'encrypt':
64.             num = num + key
65.         elif mode == 'decrypt':
66.             num = num - key

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